

Final Technical Report

Empirical model selection, persistence evaluation, appearance ReID experiment, and presentation-ready results

SELECTED PIPELINE

YOLO11n at 1280 px > tuned BoT-SORT > sparse optical-flow camera-motion compensation > stable-track presentation filter

81.0%

MOTA

72.5%

IDF1

94.1%

Precision

88.0%

Recall



Figure 1. Selected tracker at 30 seconds. The presentation overlay shows only stable tracks; the analysis CSV retains every tracker output.

Decision: Tuned BoT-SORT is the best tested configuration. Appearance embeddings and triplet-loss linking reduced identity accuracy, so both are retained only as optional review aids.

1. Project objective and completed scope

The pipeline detects road vehicles, follows them across video frames, assigns temporary track identities, measures traffic activity, and presents evidence that the system works. Track IDs represent continuity hypotheses within the video; they are not guaranteed legal or physical identities and do not require license-plate recognition.

Completed work

Requirement	Status	Completed evidence
Frame extraction	Complete	Full frame-level analysis plus review images
Vehicle detection	Complete	YOLO outputs, CSVs, counts, charts, and annotated frames
Labels	Pilot complete	96 training frames, 24 validation frames, 300 tracking frames
Persistence	Complete for clip	Six tracker families and four tuned variants compared
Appearance / ReID	Evaluated	Online ReID and custom triplet linking tested
Statistics	Complete	Counts over time, detector metrics, track-duration metrics
Visual demonstration	Complete	Final 60-second demo and synchronized comparison

Source and analysis coverage

The source video contains 53,955 frames over approximately 30 minutes. A full-video detection pass produced 387,267 car detections, with a corrected mean of 7.18 cars per frame, median 7, and maximum 28. Bounded review sequences were used for model and tracker selection so that alternatives could be compared under identical conditions.

Evaluation principle

Every tracker benchmark used the same YOLO11n weights, 1280-pixel input, 0.10 detector confidence, vehicle classes, and approved 300-frame sequence. IDF1 was prioritized for identity consistency and MOTA for combined tracking accuracy, with identity switches and fragmentation used as diagnostic evidence.

Reference limitation

The detection labels began as YOLO11n predictions, and the tracking reference began as default BoT-SORT output before manual approval. These references support a practical decision for this video, but they can favor the originating models. An independently redrawn, untouched test sequence is required before making a publication-grade superiority claim.

2. Detector selection

Four detectors were evaluated against the approved validation frames. YOLO11n provided the strongest agreement with the accepted labels and substantially lower computation time than the larger models.

Detector	AP50	Precision	Recall	F1	Mean inference
YOLO11n	1.000	1.000	0.764	0.866	189 ms/frame
YOLO11s	0.843	0.867	0.720	0.787	563 ms/frame
YOLO26n	0.826	0.909	0.587	0.713	179 ms/frame
YOLO26s	0.802	0.852	0.657	0.742	467 ms/frame

Detector comparison on approved validation frames

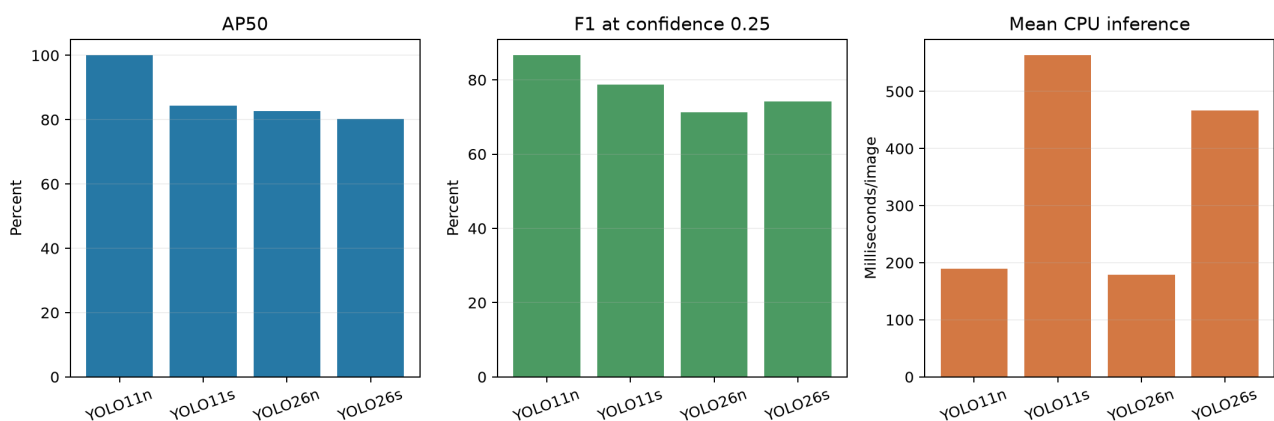


Figure 2. Detector comparison on the approved validation frames.

Selected detector

YOLO11n at 1280 pixels was retained. YOLO26n was slightly faster in this CPU run but lost substantial recall and F1. The detector comparison should be interpreted as consistency with the approved labels because YOLO11n generated their initial proposals.

Operational configuration

Setting	Value	Purpose
Image size	1280 px	Preserves distant vehicle detail
Confidence	0.10	Keeps weak boxes available for tracker recovery
Classes	2, 3, 5, 7	Car, motorcycle, bus, truck
Model	YOLO11n	Best measured detector tradeoff

3. Tracker selection

The evaluation compared ByteTrack, BoT-SORT, TrackTrack, FastTracker, OC-SORT, and Deep OC-SORT, plus tuned versions of four trackers. Tuned BoT-SORT achieved the best combined identity and tracking result.

Tracker	IDF1	MOTA	Recall	Precision	Switches	Fragments
Tuned BoT-SORT	72.5%	81.0%	88.0%	94.1%	70	47
Default BoT-SORT	72.3%	80.9%	87.3%	94.9%	77	67
ByteTrack	68.4%	77.2%	83.3%	94.9%	76	66
Tuned FastTracker	68.2%	74.3%	85.8%	89.3%	53	76
FastTracker	67.0%	75.0%	87.3%	89.1%	73	97
Tuned TrackTrack	65.5%	63.2%	91.7%	77.3%	71	53
Tuned Deep OC-SORT	63.2%	76.9%	89.0%	90.7%	133	67
OC-SORT	60.2%	68.9%	73.3%	97.3%	106	166
Deep OC-SORT	59.6%	65.1%	68.2%	98.2%	86	181
TrackTrack	43.6%	29.0%	29.3%	99.5%	4	23

Tracker comparison on the approved 300-frame sequence

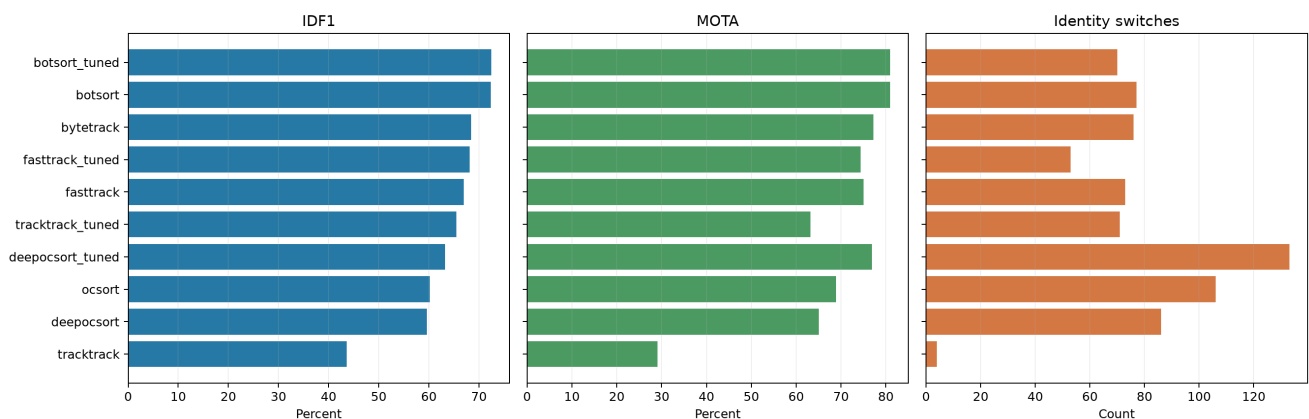


Figure 3. Tracker identity and accuracy comparison.

Why tuned BoT-SORT won

It achieved the highest IDF1 and MOTA while reducing identity switches by 9% and fragments by 30% relative to default BoT-SORT. Tuned FastTracker produced fewer switches but lower overall identity accuracy. Tuned TrackTrack recovered more vehicles but introduced too many incorrect associations.

4. Appearance embeddings and triplet-loss integration

Appearance was tested in two forms. Online BoT-SORT ReID used visual features during frame-to-frame association. A custom 128-dimensional triplet projection encoded up to eight crops per track and linked non-overlapping fragments only after class, time, motion, size, and appearance gates agreed.

Variant	IDF1	MOTA	Precision	Recall	Switches	Fragments
No ReID baseline	72.5%	81.0%	94.1%	88.0%	70	47
Triplet linker, strict 0.90	71.0%	81.0%	94.1%	88.0%	70	47
Triplet linker, 0.82	69.6%	81.0%	94.1%	88.0%	70	47
Online ReID, 0.80	68.8%	79.0%	90.1%	90.6%	77	63
Online ReID, 0.95	68.2%	80.3%	91.8%	90.1%	79	57
Online ReID, 0.90	66.6%	79.8%	91.4%	90.2%	88	58
Triplet linker, calibrated	60.5%	81.0%	94.1%	88.0%	70	47

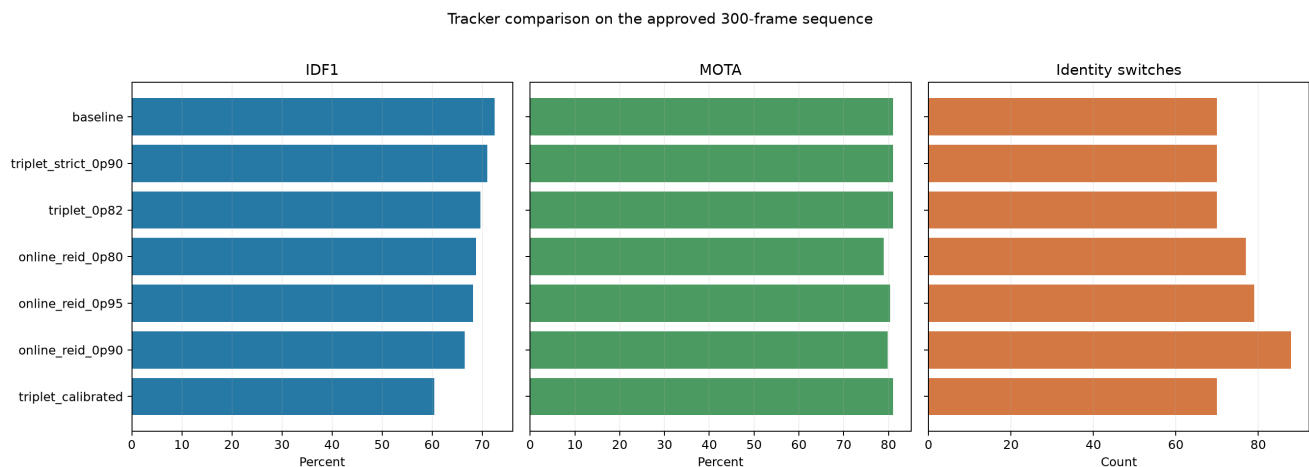


Figure 4. Every automatic appearance-assisted variant scored below the no-ReID tracker.

Finding

The triplet model improved sampled triplet ordering from 80.0% to 88.2% on average, with a best run of 90.8%. That ranking result did not transfer to safe open-scene identity merging. At its calibrated cosine threshold, the linker accepted 27 merges and reduced IDF1 by 12.0 percentage points. Even the strictest tested threshold reduced IDF1 by 1.5 points.

Decision

Automatic appearance ReID remains disabled. The learned embedding is retained only to rank suspicious track fragments for optional manual review.

5. Final configuration and full-clip result

The final tracker combines a longer persistence buffer with camera-motion compensation and conservative new-track creation.

Parameter	Selected value	Reason
High association threshold	0.25	Primary confident matches
Low recovery threshold	0.10	Recover weak distant detections
New-track threshold	0.32	Suppress weak new identities
Track buffer	60 frames	Survive about two seconds of occlusion
Match threshold	0.85	Strong geometric association
Camera compensation	Sparse optical flow	Account for camera movement
Appearance ReID	Disabled	All tested variants reduced IDF1

Full 60-second clip

Result	ByteTrack	Default BoT-SORT	Tuned BoT-SORT
Raw identity hypotheses	582	441	285
Assigned detections	20,432	23,050	22,648
Median observed duration	0.40 s	0.70 s	1.07 s
Tracks under one second	427	263	137
Maximum track span	-	49.15 s	49.15 s

Compared with default BoT-SORT, the selected tuning reduced raw identity hypotheses by 35.4%, reduced sub-second tracks by 47.9%, and increased median observed duration by 52.4%. The approved benchmark also showed a small IDF1 improvement, which indicates that the reduction was not caused only by aggressive merging.

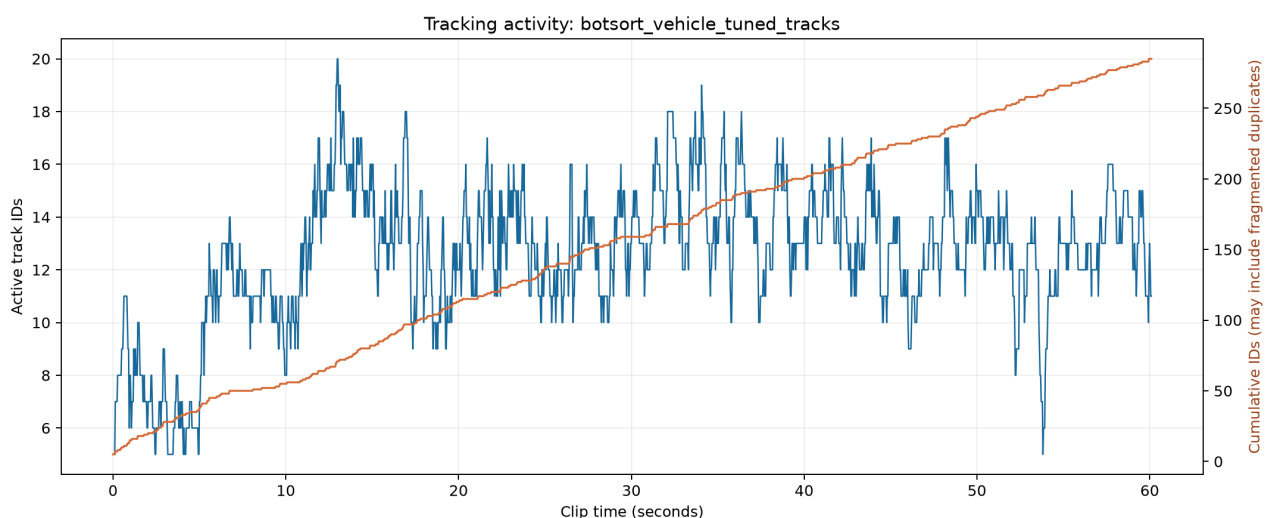


Figure 5. Tracking activity and track-duration diagnostics on the selected full-clip result.

6. Interpretation, limitations, and submission contents

What the IDs mean

The tracker assigns a new ID when a vehicle appears, predicts its next location, and associates nearby compatible detections. License-plate recognition is not required. Short occlusions are handled through motion and the 60-frame buffer. If a visually ambiguous car disappears for a long time and later returns, its exact physical identity may not be provable from this video alone.

Main limitations

Limitation	Effect	Recommended research step
Reference labels originated from tested models	Possible evaluation bias	Independently redraw an untouched test sequence
Only 420 ReID crops	Weak appearance generalization	Collect more hard negatives and fragmented identities
Moving camera and distant vehicles	Low-detail crops and viewpoint changes	Add difficult examples across scale and occlusion
Single video	Unknown generalization	Evaluate on a second route and different conditions

Submission package

The submission package includes this report, a start-here guide, supporting Markdown reports, the selected 60-second demo, a synchronized tracker-comparison video, metrics CSVs, charts, snapshots, the complete selected track CSV, configuration files, reproducible scripts, and pinned package requirements. The 7.8 GB source video and local virtual environment are excluded to keep the package portable.

Conclusion

The current project milestone is complete. The evidence supports YOLO11n at 1280 pixels plus tuned BoT-SORT with sparse optical-flow camera-motion compensation. Appearance ReID and triplet-loss merging were implemented and tested, but both produced worse identity results, so they are excluded from automatic tracking.

Method references

Ultralytics tracking documentation: <https://docs.ultralytics.com/modes/track/>

BoT-SORT: <https://arxiv.org/abs/2206.14651>

ByteTrack: <https://arxiv.org/abs/2110.06864>

Deep OC-SORT: <https://arxiv.org/abs/2302.11813>